

## Flavonoids and cancer prevention: a review of the evidence.

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The objective of this work is to review data from epidemiological and preclinical studies addressing the potential benefits of diets based on flavonoids for cancer prevention. Flavonoids are subdivided into subclasses including flavonols, flavones, flavanones, flavan-3-ols, anthocyanidins, and isoflavones. Epidemiological studies suggest dietary intake of flavonoids may reduce the risk of tumors of the breast, colon, lung, prostate, and pancreas. However, some studies have reported inconclusive or even harmful associations. A major challenge in the interpretation of epidemiological studies is that most of the data originate from case-control studies and retrospective acquisition of flavonoid intake. Differences in agricultural, sociodemographics, and lifestyle factors contribute to the heterogeneity in the intake of flavonoids among populations residing in the United States, Europe, and Asia. Dose and timing of exposure may influence the anticancer response to flavonoid-rich diets. A limited number of intervention trials of flavonoids have documented cancer preventative effects. Proposed anticancer mechanisms for flavonoids are inhibition of proliferation, inflammation, invasion, metastasis, and activation of apoptosis. Prospective studies with larger sample sizes are needed to develop biomarkers of flavonoid intake and effect. Mechanistic studies are needed to ascertain how flavonoid-rich diets influence gene regulation for cancer prevention.